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Mitigating Barren Plateaus in Quantum Neural Networks:

Optimizing the Hybrid Quantum-Classical Training Loop

Towards efficient training for scalable quantum machine learning

Student Research Poster • IIIT Delhi Quantum Centre Visit

1. INTRODUCTION

Quantum Neural Networks (QNNs)

QNNs combine parameterized quantum circuits with classical machine learning, using superposition, entanglement and interference to learn patterns in data.

Q

SUPERPOSITION

Multiple states

Q

ENTANGLEMENT

Correlated qubits

Q

INTERFERENCE

Amplify answers

Challenge → training can stall in barren plateaus.

2. THE BARREN PLATEAU PROBLEM

What is a barren plateau?

A region of parameter space where the loss landscape becomes extremely flat and gradients approach zero as system size or circuit depth grows.



Parameter landscape

Flat landscape → vanishing gradients → slow / failed training

3. WHY DOES IT HAPPEN?

1 **INCREASING CIRCUIT DEPTH**

More layers → more entanglement

2 **STRONG ENTANGLEMENT**

Sensitivity spreads across qubits

3 **RANDOM INITIALIZATION**

Unstructured parameter landscapes

4 **MEASUREMENT NOISE**

Adds uncertainty to gradients

4. THE HYBRID QUANTUM–CLASSICAL TRAINING LOOP

Quantum circuits generate outputs; classical optimization updates the trainable parameters.

Input Data	Parameterized Quantum Circuit	Measurement	Loss Function	Classical Optimizer
$\|x\rangle$	$U(\theta)$	$p(y\|\theta)$	$\mathcal{L}(\theta)$	Adam / SPSA

$$\theta \leftarrow \theta - \eta \nabla \mathcal{L}$$

The objective: keep gradients informative while controlling depth, entanglement and noise.

5. MITIGATION STRATEGIES

1 **SMART INITIALIZATION**

Problem-aware / heuristic starts

2 **LAYER-WISE TRAINING**

Increase depth progressively

3 **STRUCTURED ANSATZ**

Avoid unnecessary entanglement

4 **LOCAL COST FUNCTIONS**

Improve local gradient signals

5 **ADAPTIVE OPTIMIZATION**

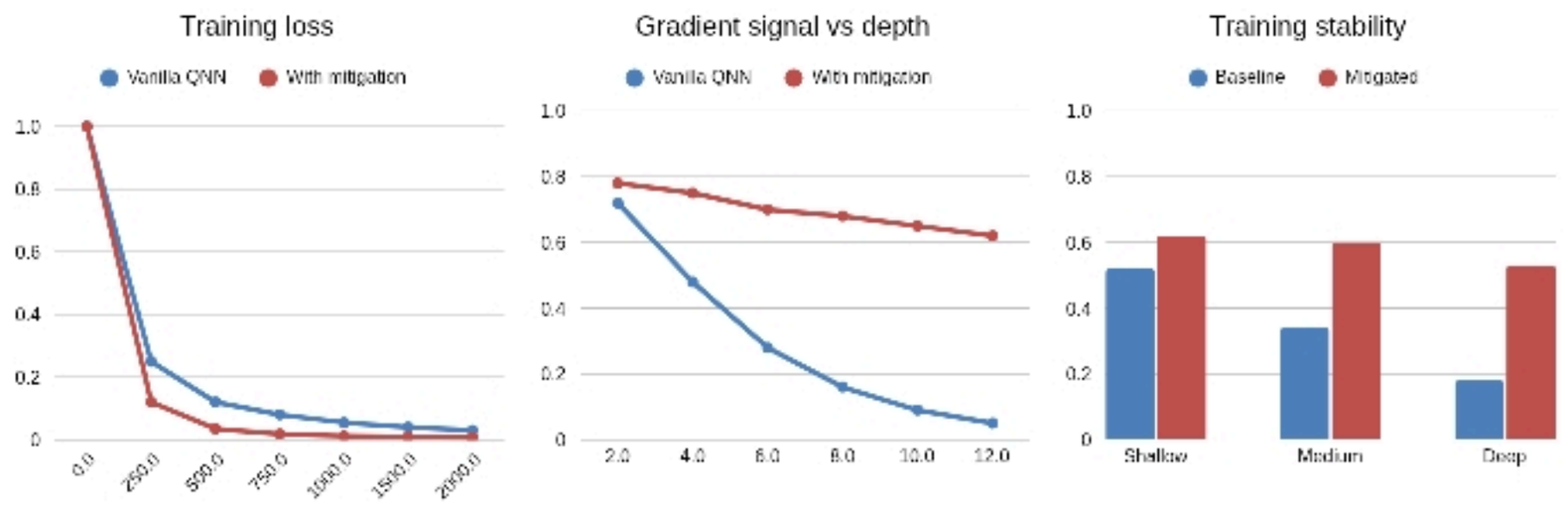
Adam, SPSA, second-order methods

6 **NOISE-AWARE TRAINING**

Calibration + realistic noise models

6. RESULTS & OBSERVATIONS (ILLUSTRATIVE)

Expected trends from mitigation-focused training designs; not presented as measured IIIT-Delhi data.



7. KEY TAKEAWAYS

✓ **Barren plateaus**

A major optimization barrier in QNN training.

✓ **Hybrid loop**

Quantum evaluation + classical updates must be co-designed.

✓ **Mitigation stack**

Initialization, layer-wise training, ansatz design and local costs work together.

✓ **Scalability**

The goal is stable gradients without sacrificing useful quantum structure.

8. FUTURE SCOPE

LARGER QUBIT SYSTEMS

Deeper circuits with mitigation

REAL HARDWARE

Noise-aware experiments

REAL-WORLD APPLICATIONS

Optimization, chemistry, finance

FAULT-TOLERANT PATH

Error correction + robust training

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DOIs: 10.1038/s41467-018-07090-4 • 10.1038/s41467-021-21728-w • 10.22331/q-2019-12-05-214 • 10.1103/PhysRevA.99.032331

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